



Stony Brook Institute
at Anhui University

安徽大学纽约石溪学院

YJ40211 Legal, Social and Ethical Issues in
Information Systems (SBU's ISE 312)
Course Information

Ch 4: Intellectual Property

4.1 Principles, Laws, and Cases

4.2 Responses to Copyright

Infringement

4.2.1 Defensive and Aggressive Responses From the Content Industries

4.2.2 The Digital Millennium Copyright Act: Anticircumvention

4.2.3 The Digital Millennium Copyright Act: Safe Harbor

4.3 Search Engines and Online Libraries

4.4 Free Software

4.5 Patents for Inventions in Software



4.1. Principles, Laws, and Cases

*"The Congress shall have Power To . . . **promote the Progress of Science and useful Arts**, by **securing for limited Times to Authors and Inventors the exclusive Right** to their respective Writings and Discoveries..."*

—U.S. Constitution, Article I, Section 8, Clause 8

Principles, Laws, and Cases

What is Intellectual Property?

- The **intangible creative work**, not its particular physical form
- Value of intelligence and artistic work comes from creativity, ideas, research, skills, labor, non-material efforts and attributes the creator provides
- Protected by copyright and patent law
 - Copyright is a legal concept that defines rights to certain kinds of intellectual property
 - Copyright protects creative works such as books, articles, plays, songs (both music and lyrics), works of art, movies, software, and videos
 - Facts, ideas, concepts, processes, and methods of operation are not copyrightable

Principles, Laws, and Cases

What is Intellectual Property?

- **Patents protect inventions**
 - An invention of any new, useful, and non-obvious process, machine, article of manufacture, or composition of matter, or any new and useful improvement thereof 在其中.
 - Patents protect the idea.
 - A patent is a set of exclusive rights granted by a sovereign state to an inventor or assignee for a limited period of time in exchange for detailed public **disclosure** of an invention.
 - Patentability requirements: novelty, usefulness, and non-obviousness.
 - The exclusive right granted to a patentee is the **right to prevent others** from commercially making, using, selling, importing, or distributing a patented invention without permission
 - Since The Venetian Patent Statute of March 19, 1474

Principles, Laws, and Cases



Trademark



Registered Trademark

What is Intellectual Property?

- **Trademark**: is a recognizable name, word, sign (logo), design, or expression which identifies products or services of a particular source from those of others
 - Protects both manifestation and idea.
- Trade secret **商业秘密**: a secret device or technique used by a company in manufacturing its products.
 - Can be a formula, practice, process, design, instrument, pattern, commercial method, or compilation of information not generally known or reasonably ascertainable by others by which a business can obtain an economic advantage over competitors or customers

Principles, Laws, and Cases

Intellectual property protection

- Protects the **intangible creative work**
 - When we buy a novel, we are buying a physical collection of paper and ink or an electronic-book file.
 - **We are not buying the intellectual property - that is, the plot, the organization of ideas, the presentation, the characters, and the events that form the abstraction**
 - that is the intangible “book”
 - We are buying the right to watch it
 - We may not make copies
 - same applies for music, video, software
 - We don't have the right to play it in a public venue or charge a fee.

Principles, Laws, and Cases

Intellectual property protection

- Protects the right of artists, authors, and inventors to gain compensation for what they create
 - The value of a book is much more than printing it
 - The value of a painting is higher than the cost of the canvas and paint used to create it
 - The value of intellectual and artistic works comes from the creativity, ideas, research, skills, and labor that the creators provided
 - Our property rights to the physical property we buy includes using it

Principles, Laws, and Cases

Discussion Question

- *How is intellectual property different than physical property?*

Principles, Laws, and Cases

Intellectual property protection

- Copyrights last for a limited time—for example, the lifetime of the author plus 70 years
- U.S. copyright law (Title 17 of the U.S. Code) gives the copyright holder the following exclusive rights:
 - To make copies of the work
 - To produce derivative works, such as translations into other languages or movies based on books
 - To distribute copies
 - To perform the work in public (e.g., music, plays)
 - To display the work in public

Principles, Laws, and Cases

Challenges of New Technology

"New technologies have been disrupting existing equilibria for centuries, yet balanced solutions have been found before."

—Pamela Samuelson,
Berkeley Law

Berkeley
UNIVERSITY OF CALIFORNIA



Principles, Laws, and Cases

Challenges of New Technology

- Digital technology and the Internet make copyright infringement easier and cheaper.
 - Photocopiers made copying of printed material easy
 - New compression technologies make copying large files (e.g. graphics, video and audio files) feasible.
 - Search engines make finding material easier.
 - Peer-to-peer technology makes transferring and sharing files easier.



Principles, Laws, and Cases

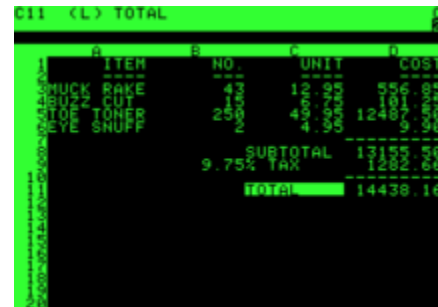
Challenges of New Technology (cont.)

- Miniaturization of cameras and other equipment enable audience members to record and transmit events.
- Scanners allow us to change the media of a copyrighted work, converting printed text, photos, and artwork to electronic form.
- New tools allow us to modify graphics, video and audio files to make derivative works.

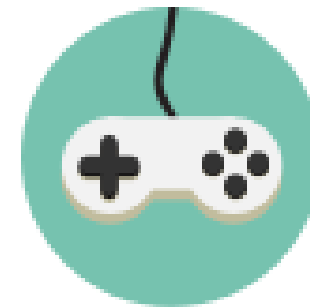


Principles, Laws, and Cases

- The first category of intellectual property to face significant threats from digital media was computer software itself
 - word processing programs
 - spreadsheet programs
 - operating systems
 - utilities
 - games
- Copying software used to be common practice: “once considered a standard and acceptable practice (if it were considered at all)”
 - *warez*: unauthorized copies of software
- Software publishers began using the term “*software piracy*” for high-volume, unauthorized copying of software



ITEM	NO.	UNIT	COST
BUCK RAKE	4	12	48
BUCK CUT	1	12	12
TOP TONER	25	4	100
EYE SHUFF	1	12	12
SUBTOTAL			172
9.75% TAX			16.725
TOTAL			188.725



Principles, Laws, and Cases

- The audio data compression format MP3, introduced in the mid-1990s, reduced the size of audio files by a factor of about 10–12.
 - People could download an MP3 song from the Internet in a few minutes
 - MP3 has no mechanism for preventing unlimited or unauthorized copying

Principles, Laws, and Cases

A bit of history

- 1790 first copyright law passed covered books, maps, and charts (1710 in UK)
- Copyright Act of 1909 defined an unauthorized copy as a form that could be seen and read visually
 - covered photography, also sound recordings and movies
- 1970s: software was copied
 - A company copied the software for a chess game from the ROM chip (read-only-memory) of the creator company
 - Because the ROM could not be read visually, a court held that the copy did not infringe the program's copyright

Principles, Laws, and Cases

A bit of history

- 1976 and 1980 copyright law revised to include software and databases that exhibit "authorship" (original expression of ideas), under "fair use" (see later)
 - 1976 law stated that the copy is in violation if the original can be perceived, reproduced, or otherwise communicated by or from the copy, directly or indirectly – an improvement over “seen and read visually”
- 1982 high-volume copying became a felony重罪
- 1992 making multiple copies for commercial advantage and private gain became a felony
 - >10 copies, worth >\$2,500 get up to 5 yrs in jail

Principles, Laws, and Cases

A bit of history

- The **No Electronic Theft (NET) Act of 1997** made it a **criminal offense to willfully infringe copyright** (for works with total value of more than \$1000 within a six-month period) even if there is no commercial advantage or private gain
 - The penalties can be severe

Principles, Laws, and Cases

A bit of history

- 1998 **Digital Millennium Copyright Act (DMCA)** prohibits making, distributing or using tools to circumvent technological protection systems that safeguard copyrighted works and included protection from some copyright lawsuits for Web sites where users post material
 - Safe-harbor provisions: Protects Web sites if they remove material when asked by the copyright holder, which offered protection from some copyright lawsuits for Web sites where users post materials
- 2005 Congress made it a felony to record a movie in a movie theater

Principles, Laws, and Cases

Fair Use Doctrine

- A principle in **copyright law** that aims to balance the rights of authors and publishers consistent with the following two goals:
 1. **Promoting production of useful work**: giving creators the incentive to produce original works by granting them exclusive rights to profit from their work
 2. **Encouraging the use and flow of information**: At the same time, the law seeks to ensure that information can be freely used for public benefit.
- The fair use doctrine allows limited uses of copyrighted materials:
 - **Uses that contribute to the creation of new work** (such as quoting part of a work in a review) and
 - **uses that are not likely to deprive authors or publishers of income for their work**.
 - For example, **Education** (even making multiple copies for classroom use)

Principles, Laws, and Cases

Fair Use Doctrine

- The 1976 law identifies **possible fair uses**, such as *“criticism, comment, news reporting, teaching (including multiple copies for classroom use), scholarship, or research”*

Principles, Laws, and Cases

Fair Use Doctrine

- Four factors considered
 1. Purpose and nature of use – commercial or nonprofit purposes
 2. Nature of the copyrighted work
 3. Amount and significance of portion used
 4. Effect of use on potential market or value of the copyright work (will it reduce sales of work?)
- No single factor alone determines
- Not all factors given equal weight, varies by circumstance

Principles, Laws, and Cases

Ethical arguments about copying

- Copying or distributing a song or computer program does not decrease the use and enjoyment any other person gets from his or her copy.
- Copying can decrease the amount of money that the copyright owner earns.
 - That is the aspect of the property that one can steal from the copyright holder
- The fact that some people copy for personal use and do not profit is irrelevant
 - Vandals 蓄意破坏者 do not profit financially from their action, but vandalism is unethical (and a crime) because it destroys—or reduces the value of—someone's property

Principles, Laws, and Cases

- Arguments people make in support of personal copying or posting content on the Web without authorization
 - "I cannot afford to buy the software or movie or pay the royalty for use of a song in my video" or
 - "I wouldn't buy it at the retail price (or pay the required fee) anyway. The company does not lose income"
 - There are many things that I cannot afford, but it does not justify just taking it
 - "The company is a large, wealthy corporation"
 - The size and success of the company does not justify taking software: programmers lose income
 - "Making a copy for a friend is just an act of generosity"
 - "This violation is insignificant compared to the billions of dollars lost to piracy by dishonest people making big profits"
 - "Everyone does it. You would be foolish not to"
 - The number of people doing something does not determine whether it is right.

Principles, Laws, and Cases



Significant Cases

- **Sony v. Universal City Studios (1984)**
 - The Sony case was the first case about private, noncommercial copying of copyrighted work that the Supreme Court decided
 - Two movie studios sued Sony for contributing to copyright infringement because some customers used its Betamax video cassette recording machines to record movies shown on television
 - 2 issues:
 - whether copyright owners can sue makers of copying equipment because some buyers use the equipment to infringe 违反 copyrights
 - whether recording a movie for personal use was a copyright infringement or a fair use

Principles, Laws, and Cases

Significant Cases

- Sony v. Universal City Studios (1984)
 - Arguments against fair use
 - People copied the entire work (against Fair use principle 3)
 - Movies are creative, not factual (against Fair use principle 2)

Principles, Laws, and Cases

Significant Cases

- Sony v. Universal City Studios (1984)
 - Arguments for fair use
 - The copy was for private, noncommercial use and generally was not kept after viewing (Fair use 1)
 - The movie studios could not demonstrate that they suffered any harm (Fair use 4)
 - The studios had received a substantial fee for broadcasting movies on TV, and the fee depends on having a large audience who view for free (~Fair use 2)

Principles, Laws, and Cases

Significant Cases

- Sony v. Universal City Studios (1984)
 - Supreme Court decided that the makers of a device with legitimate uses should not be penalized because some people may use it to infringe on copyright
 - Supreme Court decided copying movies for later viewing was fair use

Principles, Laws, and Cases



Significant Cases



- Reverse engineering: game machines
 - Sega Enterprises Ltd. v. Accolade Inc. (1992)
 - Accolade made videogames to run on Sega machines
 - Accolade needed to figure out how part of Sega's game-machine software worked
 - Accolade decompiled Sega's program (i.e., translated it from machine code to a form in which they could read and understand it) – this is *reverse engineering*
 - Sega sued.
 - Accolade won.
 - Accolade was making new games - fitting the purpose of fair use, that is, to encourage production of new creative work
 - Accolade's games might reduce the market for Sega's games, that was fair competition

Principles, Laws, and Cases



Significant Cases

- Reverse engineering: game machines
 - Atari Games v. Nintendo (1992)
 - the court rules that making copies of a program for reverse engineering (to learn how it works so that a company can make a compatible product) was not copyright infringement
 - It is a fair “research” use
 - Sony Computer Entertainment, Inc. v. Connectix Corporation (2000)
 - Connectix copied Sony’s PlayStation BIOS (the basic input–output system) and reverse engineered it to develop software that emulates the PlayStation console
 - Game players could then buy the Connectix program and play PlayStation games on their computers without buying the PlayStation console
 - Courts ruled that reverse engineering does not violate copyright if the intention is to make new creative works (video games), not copy the original work (the game systems)

Principles, Laws, and Cases



Significant Cases

- Sharing music: the Napster case
 - Napster opened on the Web in 1999 as a service allowing users to copy songs in MP3 files from the hard disks of other users
 - 50 million users little more than a year later
 - 100 million MP3 files were available on the service
 - 75% of college students surveyed by Webnoize used Napster
 - Metallica filed suit against Napster – followed by A&M
 - Eighteen record companies sued for copyright infringement and asked for thousands of dollars in damages for each song traded on Napster
 - The record companies won.

Principles, Laws, and Cases



Significant Cases

- Sharing music: the Napster case
 - Napster's arguments for fair use
 - The Sony decision allowed for entertainment use to be considered fair use
 - People make copies for personal, not commercial, use
 - Did not hurt industry sales because users sampled the music on Napster and bought the CD if they liked it
 - It was the same as a search engine, which is protected under the DMCA
 - They did not store any of the MP3 files
 - Their technology had substantial legitimate uses

Principles, Laws, and Cases

Significant Cases

- Sharing music: the Napster case (cont.)
 - RIAA's (Recording Industry Association of America) arguments against fair use
 - "Personal" meant very limited use, not trading with thousands of strangers
 - Songs and music are creative works and users were copying whole songs
 - Claimed Napster severely hurt sales
 - The record companies showed that the sales of singles were down 46% in 2000
 - Companies are required to make an effort to prevent copyright violations and Napster did not take sufficient steps
 - Napster was not a device or new technology and the RIAA was not seeking to ban the technology

Principles, Laws, and Cases

Significant Cases

- Sharing music: the Napster case (cont.)
 - Court ruled Napster liable in 2001 because they had the right and ability to supervise the system, including copyright infringing activities
 - Court ruled sharing music via copied MP3 files violated copyright
 - Napster faced civil suits that could have required payments of billions of dollars in damages.
 - After some ineffective attempts to remove unauthorized songs from its song lists, Napster shut down

Principles, Laws, and Cases

Significant Cases

- Sharing music: the Napster case (cont.)
 - *Was Napster responsible for the actions of its users?*